NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MID SEMESTER EXAMINATION, SEPTEMBER 2024

Sec ①

BTech: 1st Semester (CSE) (Section I, II, and III)
Course Name: Information Technology Workshop **Course Code: CS14106**
Maximum Time: 2 hours **Maximum Marks: 30**

Instructions:

Question 1 & 2 are compulsory for all the sections of CSE
Question 3 is compulsory for Section I and III only whereas Q4 is compulsory for Section II only
Sub-questions must be answered sequentially in one place
The marks, Course Outcome (CO) related to questions are mentioned on the right-hand side margin

Q.No.		Question	Marks	CO
1	a)	What fundamental set of programs control the internal operations of the computer's hardware? Give the name and explain in brief with suitable examples.	[3+3+4]	CO1
	b)	What do you call a program that performs a specialized task, such as a virus scanner, a file compression program, or a data backup program? Give the name and explain in brief with suitable examples.		
	c)	Word processing programs, spreadsheet programs, email programs, web browsers, and game programs belong to what category of software? Give the name and explain in brief with suitable examples.		
2	a)	The memory is characterized based on which key factors? Name them and explain in brief.	[2+2+3+3]	CO2
	b)	The two broad categories of memory used in the computer. What are they? Explain each of them in brief with suitable examples.		
	c)	A program has to be copied into what type of memory each time the CPU executes it? Give the name and explain in brief with suitable examples.		
	d)	When a CPU executes the instructions in a program, it is engaged in what process? Explain in brief.		
3	a)	Evaluate the differences between the third and fourth generations of computers. How did advancements in microprocessor technology change the landscape of computing?	[3+4+3]	CO1
	b)	Explain in brief the role of linkers and loaders in modern software development. How do they help make applications run better, and what challenges do developers face with dependencies and loading times?		
	c)	Why is secondary memory slower than primary memory? How does the operating system manage the interaction between primary and secondary memory?		
4	a)	Convert the binary number $(10101010111)_2$ to octal, decimal, and hexadecimal formats.	[3+2+2+3]	CO4
	b)	What is used for the fabrication of optical fibres that are used for communication?		
	c)	What is core in the optical fibre?		
	d)	What are the different ways of transmitting data. Explain each of them in brief with suitable example.		

Wish you all the best